M5A-R1: 144 MHz TRUE AM / EVALUATION SCHEMATIC / NOT RELEASED
1 W ERP sizing: 1.259 W connector carrier, 4.079 W PEP at m=0.80.
2S BOOST or 4S BUCK exclusive populations; training needs only controller 3.3 V.
BLOCKING: ST VHF matching not traced; PD54008 PowerFLAT numeric pads OPEN. NO LAYOUT.
OPEN: hot SOA, analog loop, RF AM and physical driver-fault response.
No PCB/layout/Gerbers/procurement. All physical results NOT_RUN. See docs/M5A_R1_REVIEW_DECISION.md.

01 Interface and local SI5351 source

File: 01_interface_source.kicad_sch

02 Minimal independent fail-off and captured ARM context

File: 02_interlocks.kicad_sch

02 Minimal independent fail-off and captured ARM context continued

File: 02_interlocksb1.kicad_sch

03 External protected pack and exclusive conversion populations

File: 03_input_power.kicad_sch

03 External protected pack and exclusive conversion populations continued

File: 03_input_powerb1.kicad_sch

04 2S boost and regulated MMIC rail

File: 04_boost_driver_supply.kicad_sch

05 Sine reconstruction and series AM modulator

File: 05_am.kicad_sch

05 Sine reconstruction and series AM modulator continued

File: 05_amb1.kicad_sch

06 144MHz source, driver, final matching and output filter

File: 06_normal_rf.kicad_sch

06 144MHz source, driver, final matching and output filter continued

File: 06_normal_rfb1.kicad_sch

07 Independent very-low-power sine-AM training path

File: 07_training.kicad_sch

07 Independent very-low-power sine-AM training path continued

File: 07_trainingb1.kicad_sch

08 Local voltage, current and thermal trip

File: 08_protection.kicad_sch

08 Local voltage, current and thermal trip continued

File: 08_protectionb1.kicad_sch

09 External supply declarations and measurement contract

File: 09_test_power.kicad_sch

ARDF-TX / NOT RELEASED

Sheet: /
File: RF2M.kicad_sch

Title: ARDF-TX 2 m A2A evaluation schematic

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 1/16

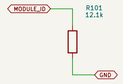
01 Interface and local Si5351 source

144.5–144.9 MHz championship; other channels only under valid country profile.
SCL activity clears ARM. Source supply/bus isolated when off. No RF claims until spectrum measurements.

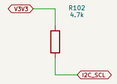
J101
Controller 20 pin



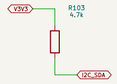
R101
12.1k



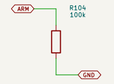
R102
4.7k



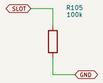
R103
4.7k



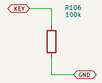
R104
100k



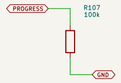
R105
100k



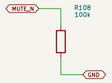
R106
100k



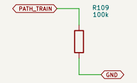
R107
100k



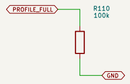
R108
100k



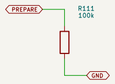
R109
100k



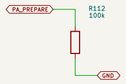
R110
100k



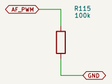
R111
100k



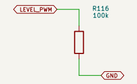
R112
100k



R115
100k



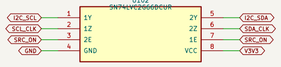
R116
100k



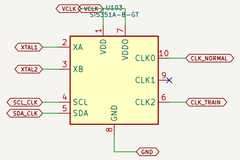
U101
TPS22919DCNR



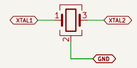
U102
SN74LVC2G66DCUR



U103
SI5351A-B-GT



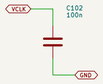
Y101
25MHz CL10p 10/10ppm



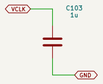
C101
100n



C102
100n



C103
1u



TPA01
CLK_NORMAL



TPA02
CLK_TRAIN



M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

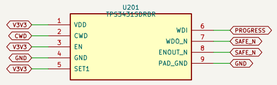
Sheet: /01 Interface and local Si5351 source/
File: 01_Interface_source.kicad_sch

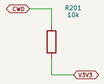
Title: 01 Interface and local Si5351 source

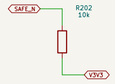
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 2/16

02 Minimal independent fail-off and captured ARM context

Raw ARM rising edge captures valid context, path and profile. ARM RC delays RF permission only.
No live path transfer. Any fault, reset, SCL-low or heartbeat expiry clears context; recovery needs a fresh ARM edge.

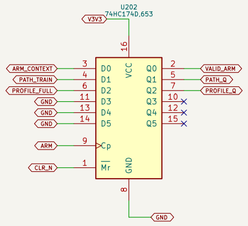














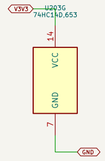


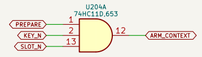




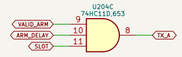


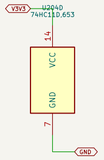


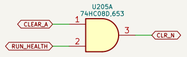


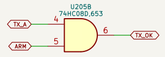


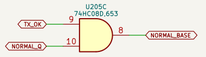


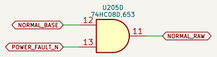


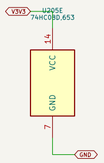


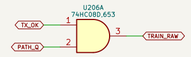


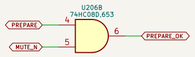


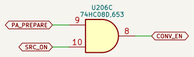


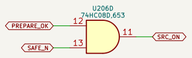


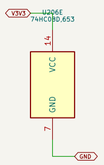


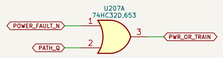




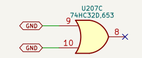












M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

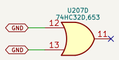
Evaluation review only; no ordering or manufacture

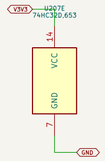
ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context/
File: 02_Interlocks.kicad_sch

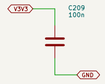
Title: 02 Minimal independent fail-off and captured ARM context

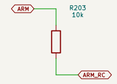
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 3/16

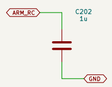


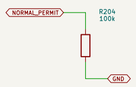


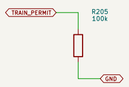


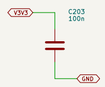


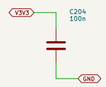


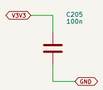


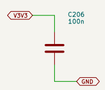


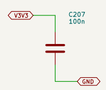


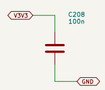


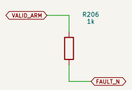
























M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /02 Minimal independent fail-off and captured ARM context continued/
File: 02_Interlocksb1.kicad_sch

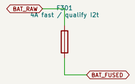
Title: 02 Minimal independent fail-off and captured ARM context continued

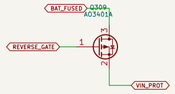
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 4/16

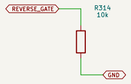
03 External protected pack and exclusive conversion populations

J301: 2S 6.0–8.4V for BOOST population OR 4S 12.0–16.8V for BUCK population.
Never fit both J302 straps. Pack/window component population is a hardware configuration, not automatic universal routing.

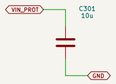


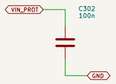


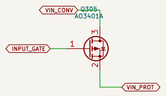


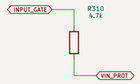


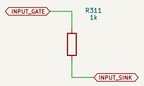


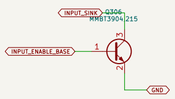


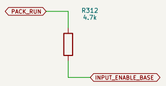


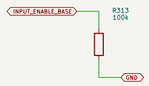




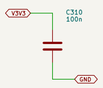




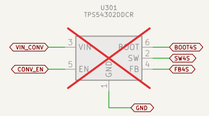


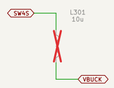








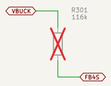






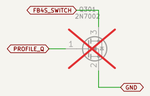


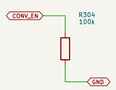


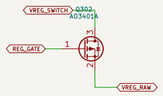


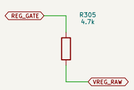












M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

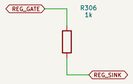
Evaluation review only; no ordering or manufacture

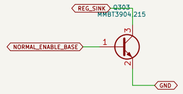
ARDF-TX

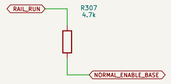
Sheet: /03 External protected pack and exclusive conversion populations/
File: 03_Input_power.kicad_sch

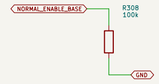
Title: 03 External protected pack and exclusive conversion populations

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 5/16

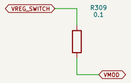






















M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

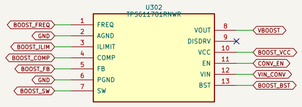
Sheet: /03 External protected pack and exclusive conversion populations continued/
File: 03_Input_powerb1.kicad_sch

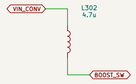
Title: 03 External protected pack and exclusive conversion populations continued

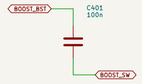
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 6/16

04 2S boost and regulated MMIC rail

BOOST or BUCK:10.0V full /7.5V reduced, captured profile. No universal buck-boost.
TPS611781 reference-derived starting compensation: must qualify loop, ripple and peak current on dummy load before PA connection.



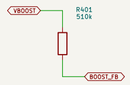


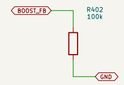


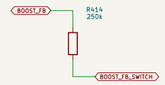


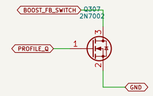


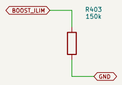


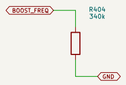


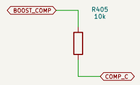


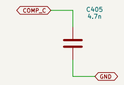


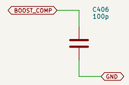


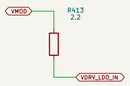


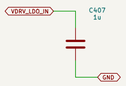




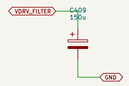




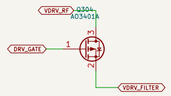


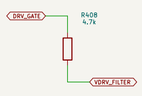


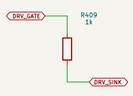


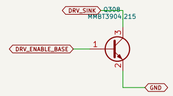


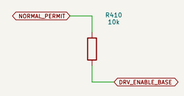


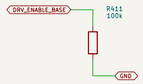


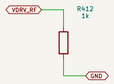
















M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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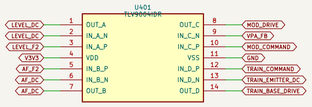
Sheet: /04 2S boost and regulated MMIC rail/
File: 04_boost_driver_supply.kicad_sch

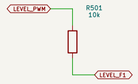
Title: 04 2S boost and regulated MMIC rail

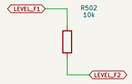
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 7/16

05 Sine reconstruction and series AM modulator

True AM: within SLOT the carrier is continuous; KEY selects filtered sine or carrier-only reference.
The controller supplies coherent LEVEL_PWM and AF_PWM (carrier offset plus sine), default800Hz. No ESP32-S3 internal DAC assumed.

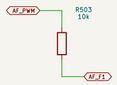


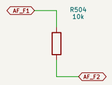




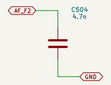


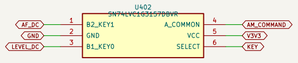


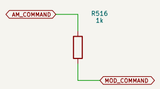


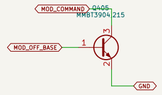


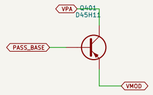


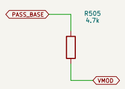


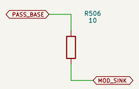


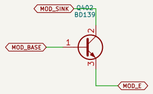


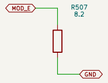


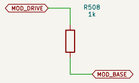


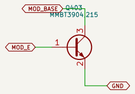


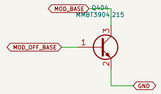


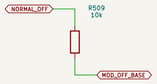


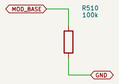


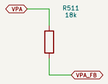


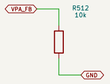


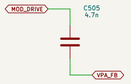




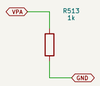


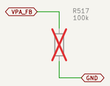


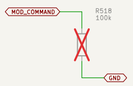


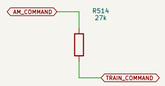












M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

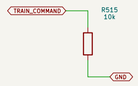
Evaluation review only; no ordering or manufacture

ARDF-TX

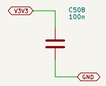
Sheet: /05 Sine reconstruction and series AM modulator/
File: 05_am.kicad_sch

Title: 05 Sine reconstruction and series AM modulator

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 8/16













M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

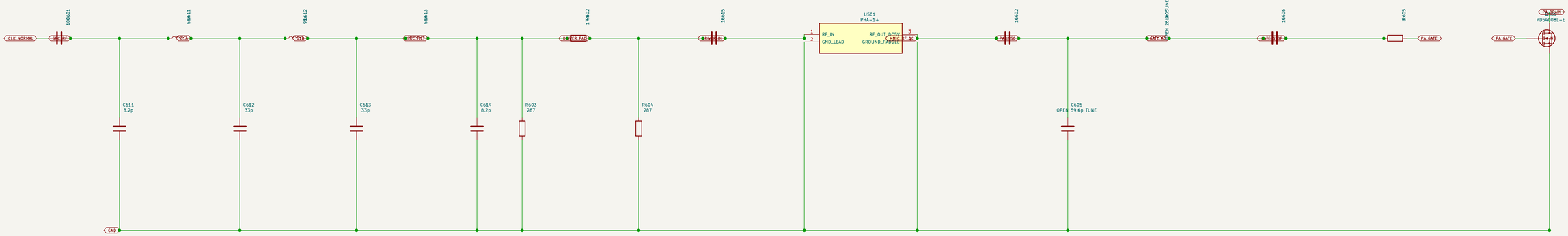
Sheet: /05 Sine reconstruction and series AM modulator continued/
File: 05_amb1.kicad_sch

Title: 05 Sine reconstruction and series AM modulator continued

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 9/16

06a Source / driver / PA, actual connected RF chain

CALCULATED EVALUATION VALUES: tune only with bounded power and calibrated loads; no144MHz hardware result exists.
Global labels connect bias, protection and supply sheets. Native symbols/pin assignments are electrically authoritative.

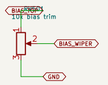








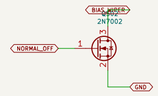






















The normal connector has no RF switch or conductor to the training connector.
Output matching OPEN: original ST drawing not traced. Do not energize the PA with this assumed-impedance harness.
LPF inductors93nH are82nH+11nH in series for current margin and obtainable values.

M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

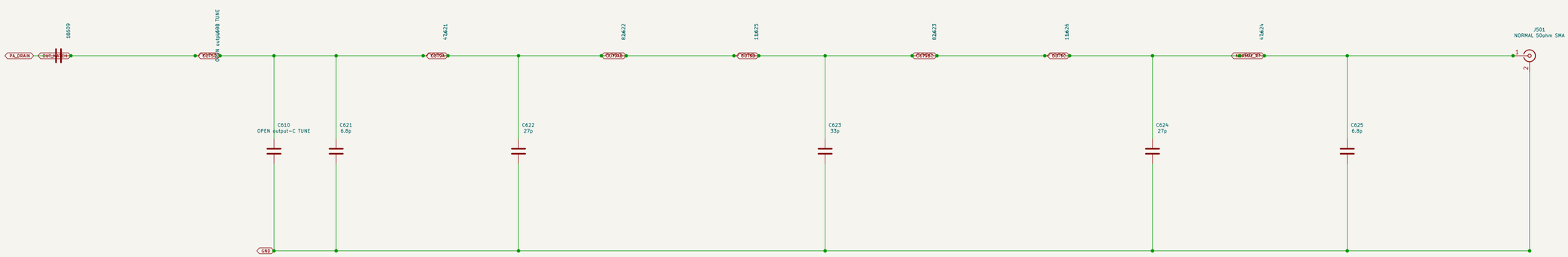
Sheet: /06 144MHz source, driver, final matching and output filter/
File: 06_normal_rf.kicad_sch

Title: 06 144MHz source, driver, final matching and output filter

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 10/16

O6b Output matching and nine-pole LPF, actual connected chain

CALCULATED EVALUATION VALUES: tune only with bounded power and calibrated loads; no144MHz hardware result exists.
Global labels connect bias, protection and supply sheets. Native symbols/pin assignments are electrically authoritative.



The normal connector has no RF switch or conductor to the training connector.
Output matching OPEN: original ST drawing not traced. Do not energize the PA with this assumed-impedance harness.
LPF inductors93nH are82nH+11nH in series for current margin and obtainable values.

M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /06 144MHz source, driver, final matching and output filter continued/
File: 06_normal_rfb1.kicad_sch

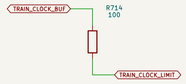
Title: 06 144MHz source, driver, final matching and output filter continued

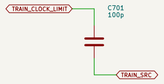
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 11/16

07 Independent very-low-power sine-AM training path

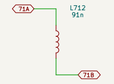
Separate source CLK2, current-modulated small RF BJT, own filter/pad and incompatible MCX port.
No connection to watt PA drain/output. Target1uW carrier, optional1-10uW via calibrated fixed pad; actual calibration and isolation OPEN.

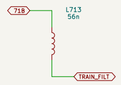


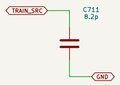


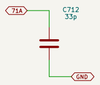


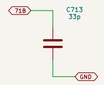




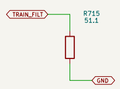


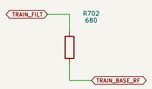


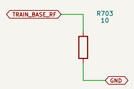


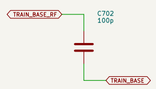


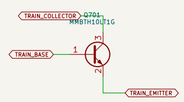


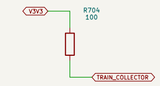


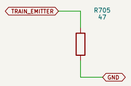


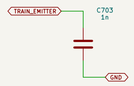


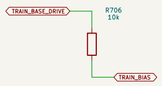


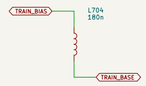


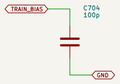


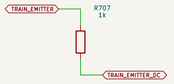


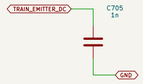


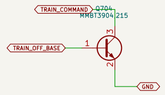


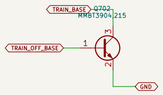


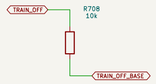


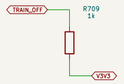


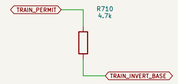


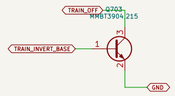


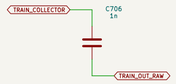








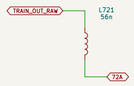


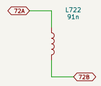


M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

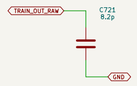
Evaluation review only; no ordering or manufacture

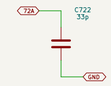
ARDF-TX		
Sheet: /07 Independent very-low-power sine-AM training path/		
File: 07_training.kicad_sch		
Title: 07 Independent very-low-power sine-AM training path		
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 12/16

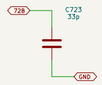


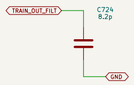


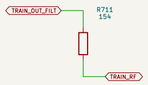


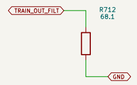


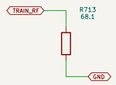


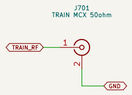
















M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /07 Independent very-low-power sine-AM training path continued/
File: 07_trainingb1.kicad_sch

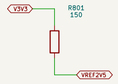
Title: 07 Independent very-low-power sine-AM training path continued

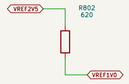
Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 13/16

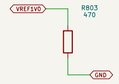
08 Local voltage, current and thermal trip

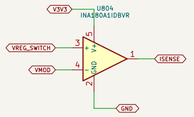
Analog faults only affect normal PA population: training uses bounded3.3V and protected host pack.
Eight comparators, one current amplifier, one common reference: distinct voltage/current/sensor functions. No permanent RF detector or ADC.

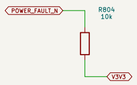


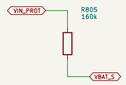


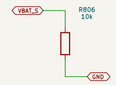


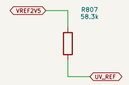


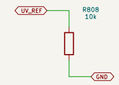


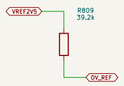


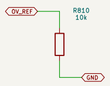


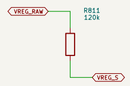


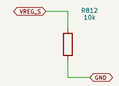


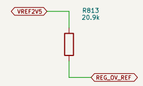


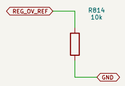


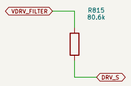


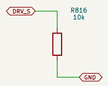


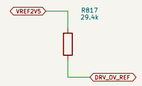


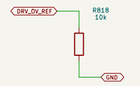


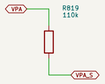


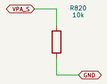


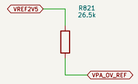


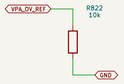


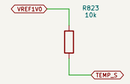


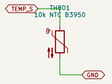


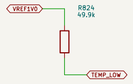


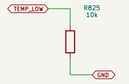


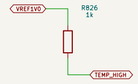


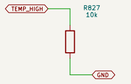












MSA-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

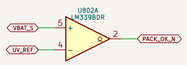
Evaluation review only; no ordering or manufacture

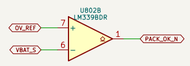
ARDF-TX

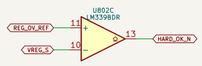
Sheet: /08 Local voltage, current and thermal trip/
File: 08_protection.kicad_sch

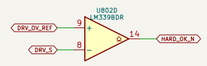
Title: 08 Local voltage, current and thermal trip

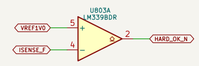
Size: A2	Date: 2026-10-15	**Rev: MSA-R1**
KiCad E.D.A. 10.0.6		Id: 14/16

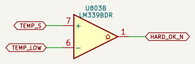


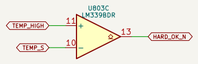


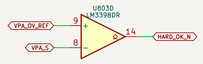


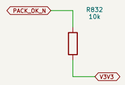


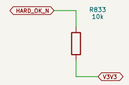








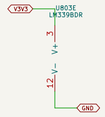


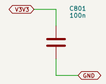


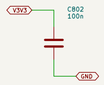


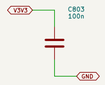


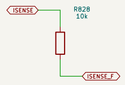


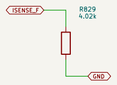


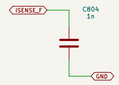






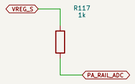


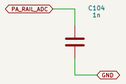














M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

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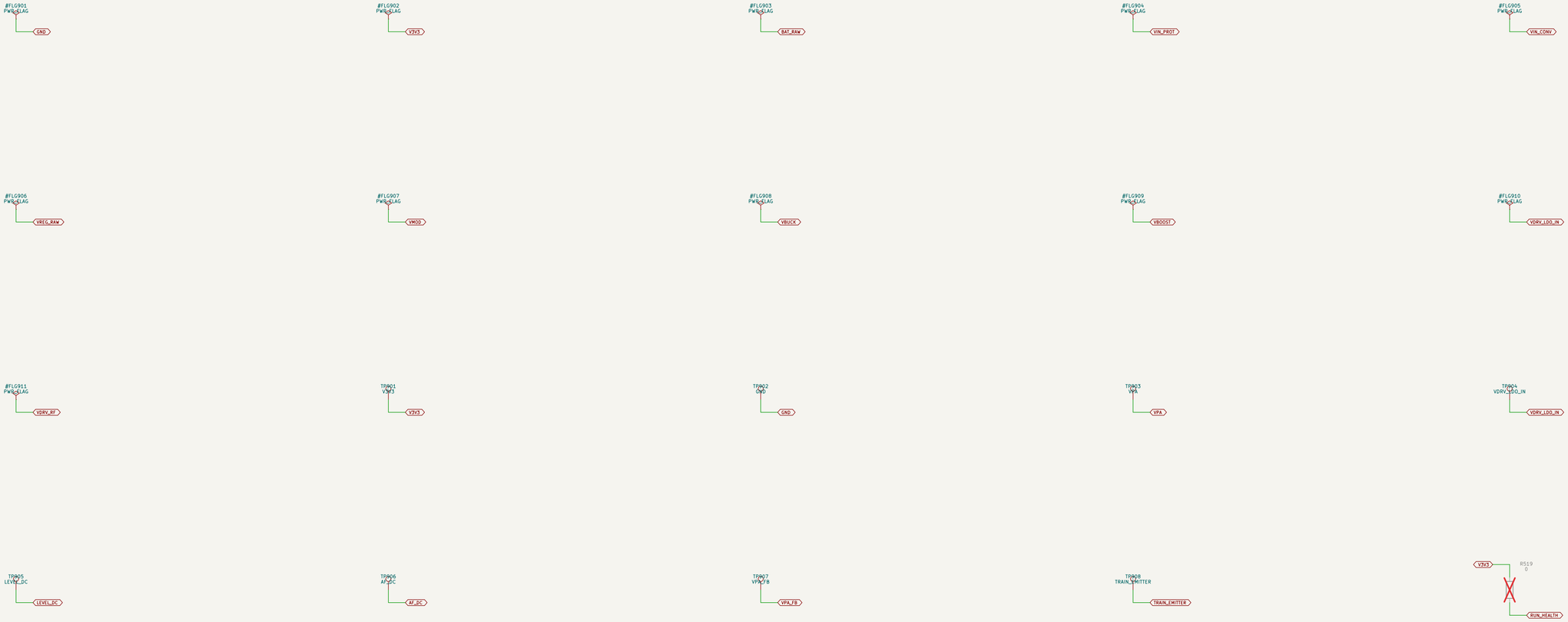
Sheet: /08 Local voltage, current and thermal trip continued/
File: 08_protectionb1.kicad_sch

Title: 08 Local voltage, current and thermal trip continued

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 15/16

09 External supply declarations and measurement contract

No RF2M PCB exists in M5A-R1. All RF/thermal/safety tests NOT_RUN.
PWR_FLAG declares a measured external source; it does not validate protection or a supply rating.
Read OPEN items in docs/M5A_R1_SCHEMATIC_REVIEW.md before an evaluation PCB.



M5A-R1: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /09 External supply declarations and measurement contract/
File: 09_test_power.kicad_sch

Title: 09 External supply declarations and measurement contract

Size: A2	Date: 2026-10-15	**Rev: M5A-R1**
KiCad E.D.A. 10.0.6		Id: 16/16